

Viewpoint-Specific Capability in Single-Camera Motion Analysis

A three-stage validation of measurability, reliability and discriminative validity: squat and bench press
MediaPipe Pose landmarks • DTW / DBA viewpoint-specific template construction

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Repetition-count accuracy 87 % 40 / 46 clips matched the manual count exactly	Drop in the target metric itself -50 pts Squat knee valgus • frontal 0°: 99.2 → 49.0 (single-metric score)	Metrics entering the weighted score 23 / 42 42 → measurable 33 → reliable 29 → scored 29; every exclusion is reported with its reason	Validated capabilities 6 / 12 viewpoint × fault cells passing discriminative validity: squat 4, bench 2
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ABSTRACT

Background Commercial home-fitness systems (e.g. **Tempo**, **Tonal**) already output form scores, but most rely on **dedicated hardware**, **depth sensing** or **multiple views** to secure stable observation, and few state the conditions under which their measurements remain valid. The user therefore cannot tell whether a low score means the **movement really was poor** or the **camera simply could not see that fault** — the latter creates a false sense of safety. This study deliberately restricts the input to a **single smartphone camera**.

Methods Because a single camera offers little observational redundancy, the pipeline is deliberately conservative. **Squat and bench press** were each recorded from three viewpoints; fixed viewpoints were **physically set and verified with a goniometer**, while the self-selected viewpoint angle was recovered post hoc by shoulder–hip projected-shortening self-calibration (47°). MediaPipe Pose supplies 33 landmarks, converted into **7 metrics** per exercise. Cycles are segmented by a hysteresis state machine with multi-signal consensus voting; after phase resampling, curves are aligned by **DTW** and averaged by **DBA** into a viewpoint-specific template, penalising only the pre-declared adverse direction. Validation proceeds through **measurability → reliability → discriminative validity**.

Results An iPhone prototype was completed. **12 template clips** (squat 6, bench 6) and **46 test clips** (squat 26 incl. 6 off-axis, bench 20) were recorded, with $n = 3$ standard and $n = 1$ induced-fault trials per cell; **exact repetition-count agreement was 87% (40/46)**. Of 42 viewpoint × metric pairs, **33 were measurable → 29 reliable → 29 scored** (squat 16/21, bench 17/21). Of 12 viewpoint × fault cells, **6 detected, 4 unmeasurable, 2 below threshold** (squat 4/1/1, bench 2/1/3); thresholds of 15/20/25 give identical verdicts.

Contribution The three-stage framework makes the system **declare the boundary of its own measurement capability first**, turning a single score into a **viewpoint-specific capability map**. **Viewpoint is the decisive factor for the capability ceiling**: knee valgus falls from ~50 at frontal 0° to ~2 at self-selected 47°, and is **unmeasurable** at sagittal 90°; squat depth passes at all three. Bench press is the mirror image — elbow flare is most sensitive overhead (~68), grip width is measurable only at the self-selected view (~80). **Limitations**: $n = 1$ induced-fault trial per cell; one participant per group.

01 Motivation

Commercial home-fitness systems (e.g. **Tempo**, **Tonal**) can indeed score movement quality, but they mostly **rely on dedicated hardware**, **3D depth sensors** or **an extra camera view** to buy stable observation conditions.

This study tightens the constraint to **one smartphone camera**. Under that hardware limit an overlooked question appears: **does a low score mean the movement was genuinely poor, or that this viewpoint simply cannot see where you went wrong?** The latter is far more dangerous — the user believes they did it right.

The objective is therefore **not scoring alone**, but two things: ① a deployable single-camera processing pipeline; ② a **measurability–reliability–discriminative-validity** three-stage validation that delimits which exercise, at which viewpoint, can reliably judge which fault — a **viewpoint-specific capability map**.

02 System Pipeline

Following from the motivation The input is a **single smartphone camera** only — no depth, no multiple views — so the rigour is placed in **processing and validation**: cycle segmentation → phase resampling → **DTW alignment** → **DBA template synthesis** → three-stage validation, ending in a **viewpoint-specific capability map**. The test exercises are **squat and bench press**, each recorded from three viewpoints.

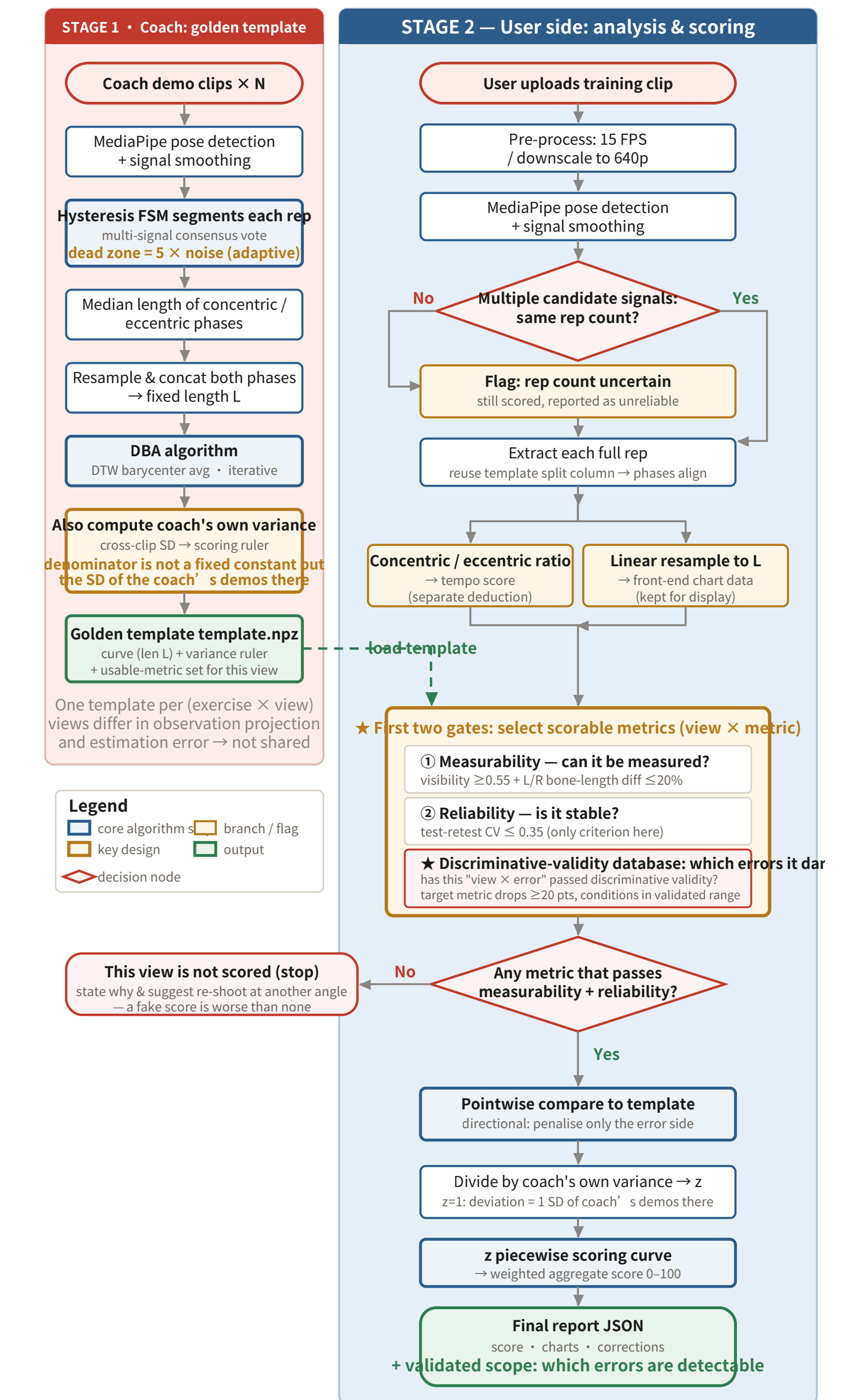


Fig. 1 Two-stage architecture. Stage 1 synthesises a golden template by DBA and uses the demonstrator's own variability as the scoring scale; Stage 2 filters scorable metrics through measurability and reliability, then applies the pre-established discriminative validity to output a score together with its validated scope of use.

03 Measurement Definitions

① Body-referenced mediolateral axis

Left–right distances in the image change with body orientation, so we first define:

$$\hat{e}_{ml} = (P_{L hip} - P_{R hip}) \text{ with the vertical component removed, divided by its own length}$$

making it **parallel to the ground** and a **unit vector**. All distances are then computed as the **projection onto $\hat{e}_{ml}$** , giving the participant's **own mediolateral width** rather than an apparent on-screen distance.

② **Normalisation to a dimensionless ratio** Absolute lengths depend on stature and camera distance, so every distance is **divided by a reference length from the same frame** (inter-ankle distance for knee valgus, thigh length for depth). The scale factor cancels on division, reducing differences due to body size and shooting distance.

08 Conclusions & Outlook

This study delivers a **deployable single-camera motion-analysis prototype** and establishes an **empirical boundary for measurement quality and fault-detection capability**. **Viewpoint is not a shooting tip; it is the experimental factor that sets the system's ceiling**: the drop in the knee-valgus metric itself goes **frontal 0° ~50 pts (detected) → self-selected 47° ~2 pts (below threshold) → sagittal 90° unmeasurable**.

Contributions ① a deployable two-stage system; ② a **ground-truth-free surrogate** for coordinate trustworthiness (bone-length L/R difference); ③ a three-stage framework that can report its own capability boundary, with the **validated scope** of use bound to image-estimated relative angle.

Limitations ① $n = 1$ induced-fault trial per cell (standard $n = 3$), so **no effect sizes or confidence intervals are estimated**; the results show a consistent trend but **still require a larger sample and independent participants**. ② One participant per group. ③ Reliability was measured on the standard group itself. ④ Angles are **2D image estimates**, and 50–90° was **not densely sampled**.

REFERENCES

- [1] Ortiz A, et al. Reliability and concurrent validity between 2D and 3D evaluations of knee valgus during drop jumps. Open Access J Sports Med. 2016;7:65–73. [2] Straub RK, Powers CM. A biomechanical review of the squat exercise. Int J Sports Phys Ther. 2024;19(4):490–501. [3] Bazarevsky V, et al. BlazePose: on-device real-time body pose tracking. arXiv:2006.10204, 2020. [4] Sakoe H, Chiba S. Dynamic programming algorithm for spoken word recognition. IEEE Trans ASSP. 1978;26(1):43–49. [5] Petitjean F, et al. A global averaging method for dynamic time warping. Pattern Recognit. 2011;44(3):678–693. [6] Savitzky A, Golay MJE. Smoothing and differentiation of data by simplified least squares procedures. Anal Chem. 1964;36(8):1627–1639. [7] Hopkins WG. Measures of reliability in sports medicine and science. Sports Med. 2000;30(1):1–15.

React • Capacitor (iOS) • FastAPI • Railway • MediaPipe Pose • SciPy • tslearn | Smoothing: Savitzky–Golay[6][3]

All four decision thresholds (0.55 / 20% / 0.35 / 20 pts) were derived in this study's pilot and declared in advance, not taken from the literature

The measurement validity of the original KASR is cited from [1]; this study uses a MediaPipe landmark-adapted formulation

03 Measurement Definitions cont.

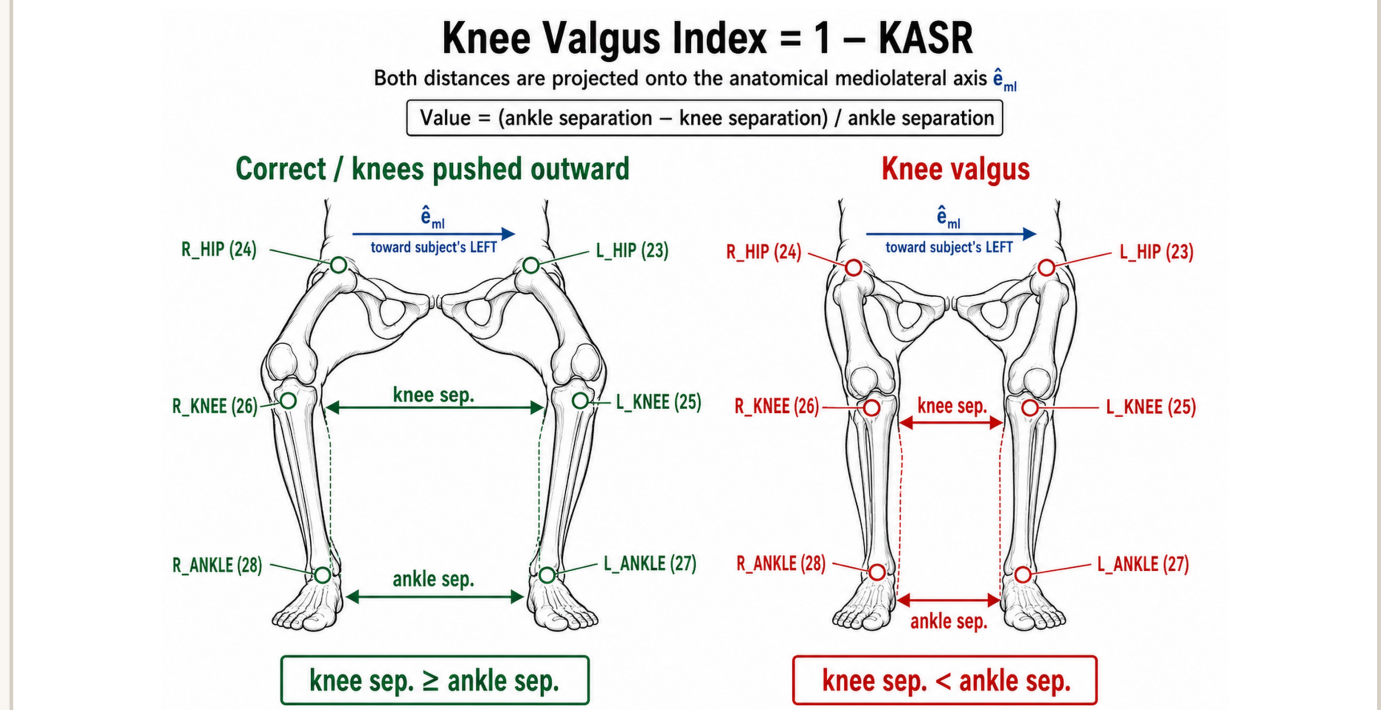


Fig. 2 Knee valgus = 1 – KASR. Following the KASR (knee-to-ankle separation ratio) concept, a corresponding dimensionless ratio is constructed from MediaPipe knee / ankle landmarks. The original KASR is measured at the lateral femoral epicondyle and lateral malleolus and shows 2D–3D agreement of ICC = 0.96[4] in a drop-jump task; here the concept is applied to the bilateral squat. **1 – KASR is a signed monotonic transform introduced in this study** (positive = knees narrower than ankles), not the published metric; both distances are measured along $\hat{e}_{ml}$. Measured at peak depth (frontal 0°): standard ~0.265, induced valgus +0.234.

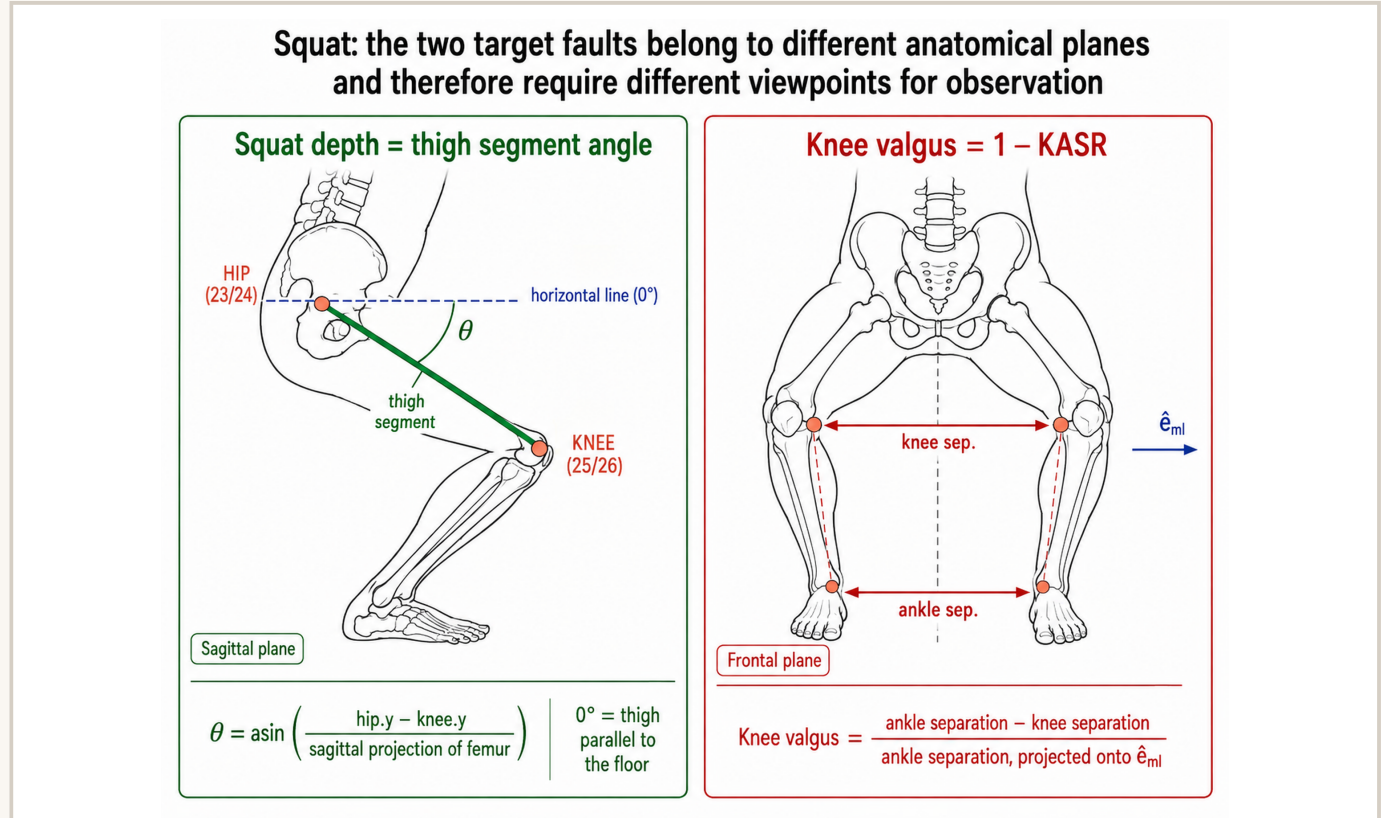


Fig. 3 Squat depth is operationally defined as the elevation angle of the thigh segment relative to the horizontal (3D segment elevation angle, not a planar projection angle). 0° corresponds to the literature boundary of "thigh parallel to the floor"[42], standing is about ~90°, and more negative means shallower. Measured unilaterally, automatically choosing the more visible side. At peak depth (sagittal 90°): standard ~9.7°, induced shallow squat ~31.8°.

04 Core Methods

① **Cycle segmentation: hysteresis state machine + consensus voting** The dead-band half-width is $5 \times$ the noise estimate (3–20% of movement range), **adapting** to the signal-to-noise ratio; the signal must cross it for an up/down transition to count, **replacing a single find_peaks** (which miscounts pauses and tremor as repetitions). The final count is then decided by **clustered voting over the counts produced by several candidate signals**, restricted to the **K most visible** of them.

② **Golden template: DTW first, then DBA** Two demonstrations differ in tempo to begin with, so index-wise subtraction mistakes "faster or slower" for "different posture". The time axes are therefore aligned by DTW[4] and then iteratively updated by DBA[5] into a **representative golden template**.

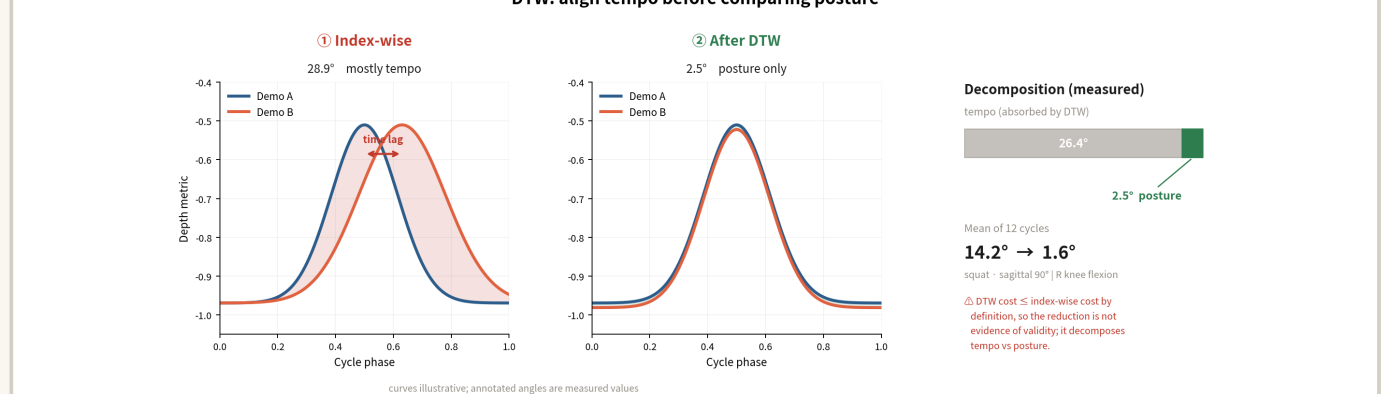


Fig. 4 DTW removes tempo from the difference. Index-wise subtraction gives 28.9°, of which 26.4° is tempo and only 2.5° is true postural difference; averaged over 12 cycles, 14.2° → 1.6°. **DTW cost is ≤ index-wise cost by definition**, so the reduction is not in itself evidence of validity — its purpose is to decompose the source of the difference.

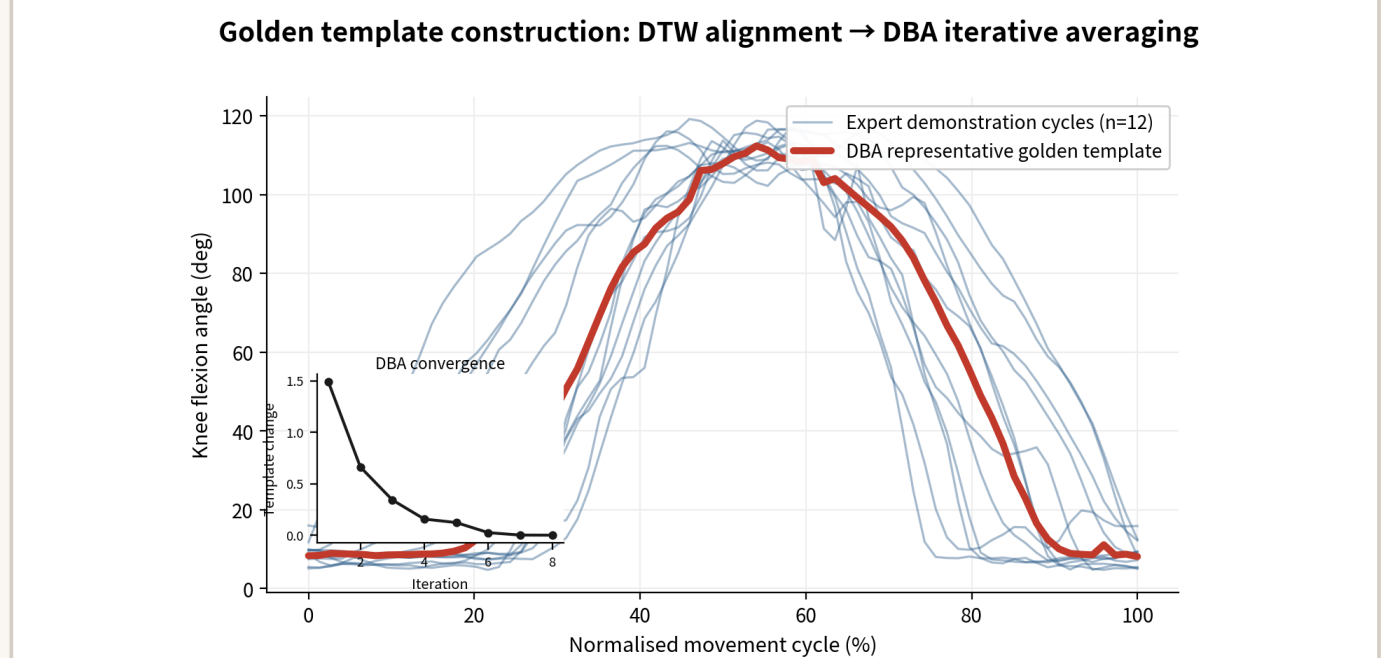


Fig. 5 Thin lines are the 12 cycles segmented from the 2 demonstration clips at this viewpoint ($n = 12$ cycles, not clips); the thick line is the representative golden template from DBA. One template is built per exercise × viewpoint **different viewpoints have different observational projections and estimation errors**, so templates are not shared.

09 Transferability

The validation framework transfers beyond this task. "Measurability–reliability–discriminative validity" is tied neither to fitness nor to MediaPipe — any system that combines **monocular vision + measurement + advice** faces the same question: on what grounds does it trust what it measured?

Transfer domain	The isomorphic problem
Rehabilitation movement assessment	Home rehab is unsupervised; "performed incorrectly yet passed" directly affects outcome
Fall-risk screening in elder care	Sit-to-stand and gait tests have clinical scales, but single-camera reliability and validity are undefined
Occupational ergonomics	Lifting-posture assessment (REBA / RULA) needs angle measurement and is still scored by hand
General vision-language models (VLM) as movement coaches	They produce confident natural-language judgements but will not say "I cannot see this from this angle!" — this study measured exactly that failure (valgus information disappears at 90°), so the framework is an evaluation protocol they need too

04 Core Methods cont.

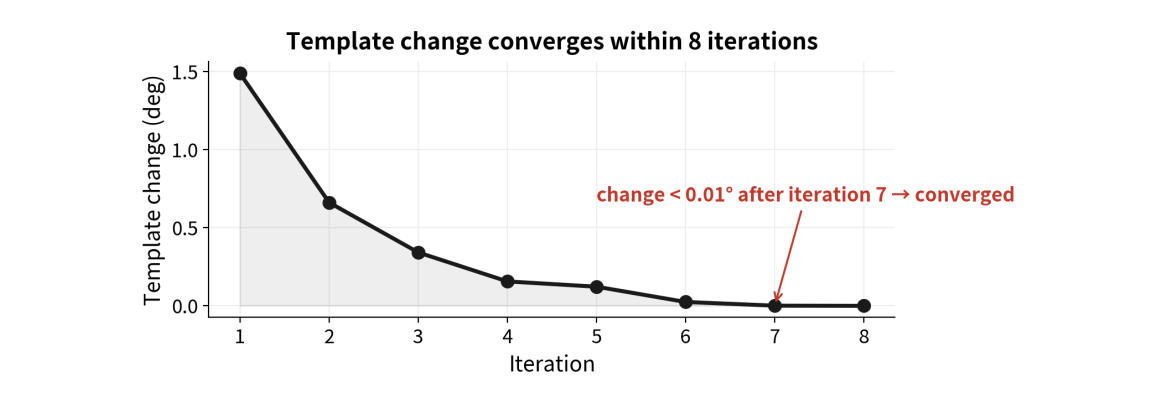


Fig. 6 DBA iterative convergence (representative example: squat × sagittal 90°). The change falls below 0.01° after the 7th iteration, so the number of iterations is fixed at 8.

③ **Directional deviation comparison** Using [user – template] amounts to "any difference from the demonstrator is penalised", yet **some deviations from the template are not faults**. Instead, an adverse direction is declared in advance for each metric, and only point-wise deviations in that direction are scored.

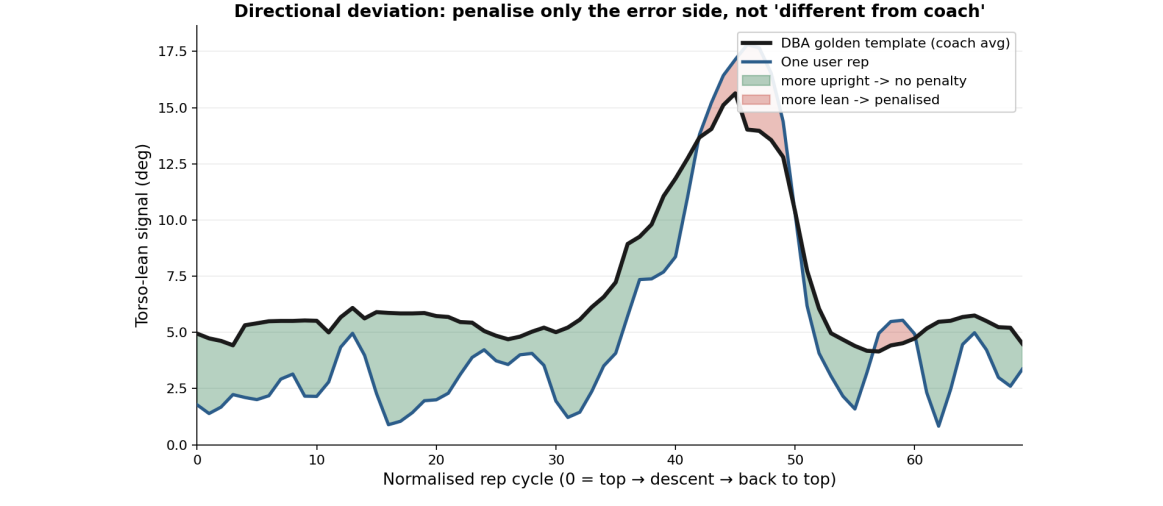


Fig. 7 Scoring direction for point-wise comparison (squat depth): only insufficient depth is defined as the penalised direction — red = shallower than the template and penalised; green = deeper than the template, not penalised on this metric.

05 Validation Design

Exercise	View 1	View 2	View 3	Target faults
Squat	Frontal 0°	Self-sel. ~47°	Sagittal 90°	Knee valgus (frontal plane) • insufficient depth (sagittal plane)

Bench	Overhead	Self-selected	Sagittal 90°	Elbow flare • narrow grip
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How viewpoint angles were set Fixed viewpoints (0° / 90° / overhead) were **physically positioned and verified with a goniometer** during recording; the self-selected viewpoint angle was recovered post hoc by self-calibration from 2D shoulder + hip projected shortening (47°, range 44–50°).

Participants 1 experienced, 1 novice
Clips 12 for template building + **46 test clips** (18 standard / 12 induced-fault / 10 novice + 6 angle sweep); the 18 standard clips also serve as the reliability calibration set

Clip set	Purpose	Maps to
Demonstration 12 clips (6 views × 2)	Segment cycles → DTW align → DBA template	Figs. 4, 5, 6
Standard 18 clips (not used for templates)	Test–retest CV → select scorable metrics	23 / 42
Per view: standard $n = 3$ vs induced fault $n = 1$	Between-condition comparison at the same view; drop in the target metric itself	Fig. 8
Knee valgus at 0° / 47° / 90° + 6 off-axis	Fault held constant, viewpoint angle varied	Fig. 8
All 46 clips	Automatic vs manual repetition count (whole-clip exact match)	87%

Two angle sources coexist Fixed viewpoints were set by **goniometer**; the angle values reported here are recovered by **self-calibration** from 2D shoulder + hip projected shortening (0° and 90° are the **calibration endpoints**). The user-placed camera fell at 47° (44–50°). For the bench press the supine posture degrades 2D angle estimation, so **discrete viewpoint validation** was used instead.

06 Three-Stage Framework

Measurability → reliability → discriminative validity. The first two gates select **scorable metrics**; the third is a fault-detection capability **established in advance during the experiment**, so the deployed app only performs a table lookup.

Exercise	All pairs	① Measurable	② Reliable	③ Scored
Squat	21	16	15	15
Bench	21	17	14	14
Total	42	33	29	29

① Can it be measured? Visibility ≥ 0.55 Bone-length L/R diff. ≤ 20%	② Is it stable? Test–retest CV ≤ 0.35	③ May it enter the total? Minus stand-alone penalties and whole-viewpoint suspension
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6 validated ③ **Discriminative validity** can the fault be detected • viewpoint × fault The target metric's own single-metric score must fall by **≥ 20 points** and the shooting condition must lie inside the validated angle range → only then is the cell listed as a validated detection capability: 6 cells (Fig. 8).

Being gated out ≠ a score of 0; it means insufficient measurement qualification, so no score is issued. Had all 42 pairs been scored, **13 would have been "not measurable / not stable, yet reported anyway"**, leaving the user unable to tell poor movement from poor measurement.

Basis of the four thresholds Visibility 0.55: below this, coordinates begin to jitter between frames; bone-length L/R difference 20%: the boundary between squat 0.7–14.1% and bench forearm 32–47%; test–retest CV 0.35: median 0.028, gated cases 0.43–9.90; target-metric drop 20 points: declared in advance, with ~15 / ~20 / ~25 giving identical verdicts.

Squat example: the same metric, a different fate at each viewpoint

Metric	Frontal 0°	Self-sel. 47°	Sagittal 90°
Knee valgus	pass	pass	gate ①
Squat depth	pass	pass	pass
Knee symmetry	pass	pass	gate ①
Torso lean	pass	gate ②	pass

Gate ① = not measurable (the required landmarks are not visible at that viewpoint) Gate ② = not stable (test–retest CV > 0.35)

07 Results

Results by exercise (total = squat + bench)

Exercise	Measurable	Reliable	Scored	Validity detected / below / unmeasurable
Squat	16 / 21	15	15	4 / 1 / 1
Bench	17 / 21	14	14	2 / 1 / 3
Total	33 / 42	29	29	6 / 2 / 4

Viewpoint-specific discriminative validity
Bars: change in the target metric's own score after inducing the fault; < 20 counts as detected

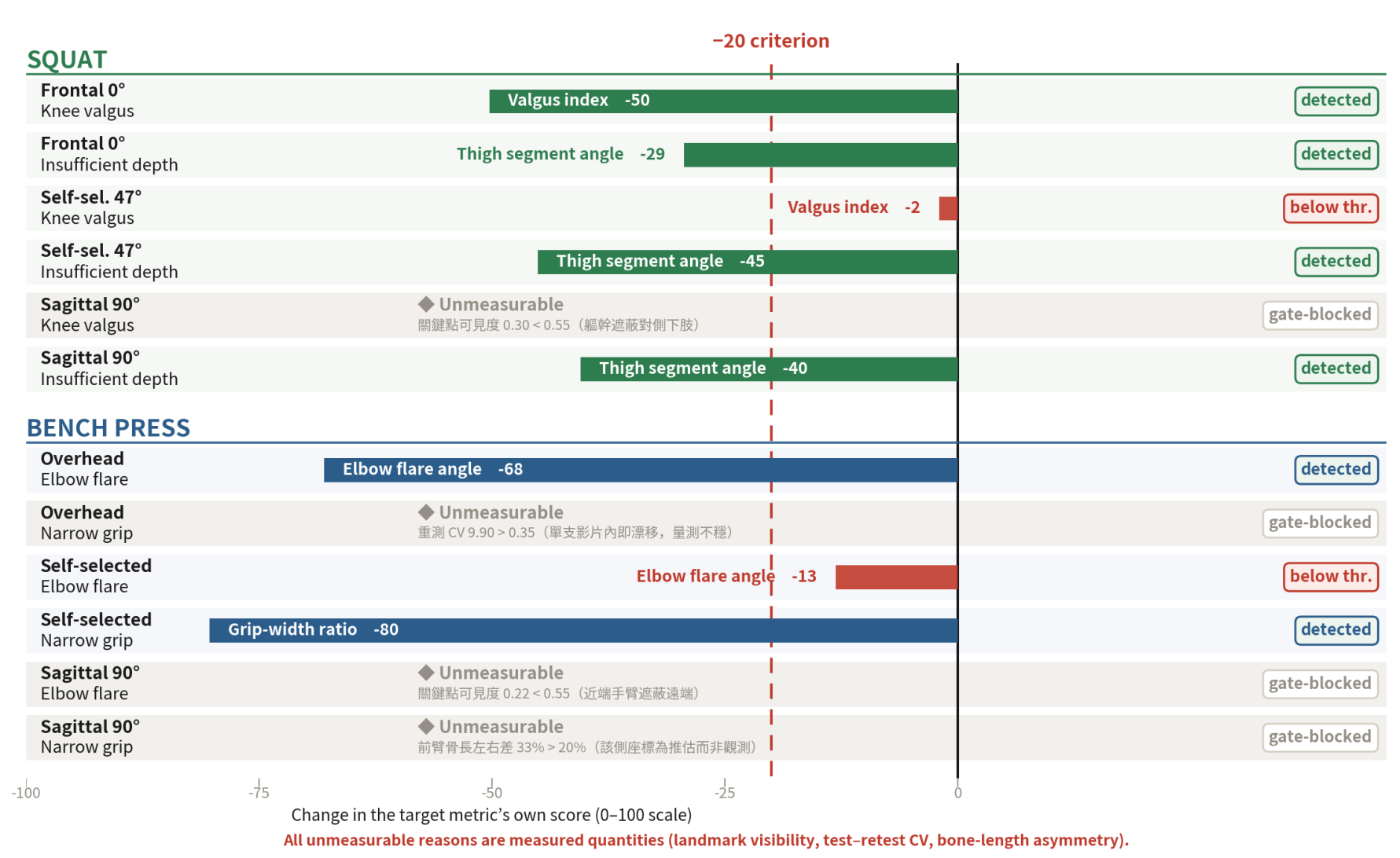


Fig. 8 Viewpoint-specific discriminative validity (12 viewpoint × induced-fault cells, judged individually). Bars show the change in the target metric's own single-metric score; a cell counts as detected only past ~20. Every "unmeasurable" verdict rests on a measured quantity (landmark visibility, test–retest CV, bone-length L/R difference), not on post hoc inference.

Why the verdict always uses the target metric itself rather than the weighted total A weighted total cannot separate "the target fault was detected" from "other metrics dragged the score down" — squat knee valgus at the self-selected view: the **weighted total falls 95 → 7 (~88)**, which looks like a very sharp detection, yet the **knee-valgus metric itself falls only 2 points (93 → 91)**. The total was pulled down by other metrics; the system never actually saw that valgus.

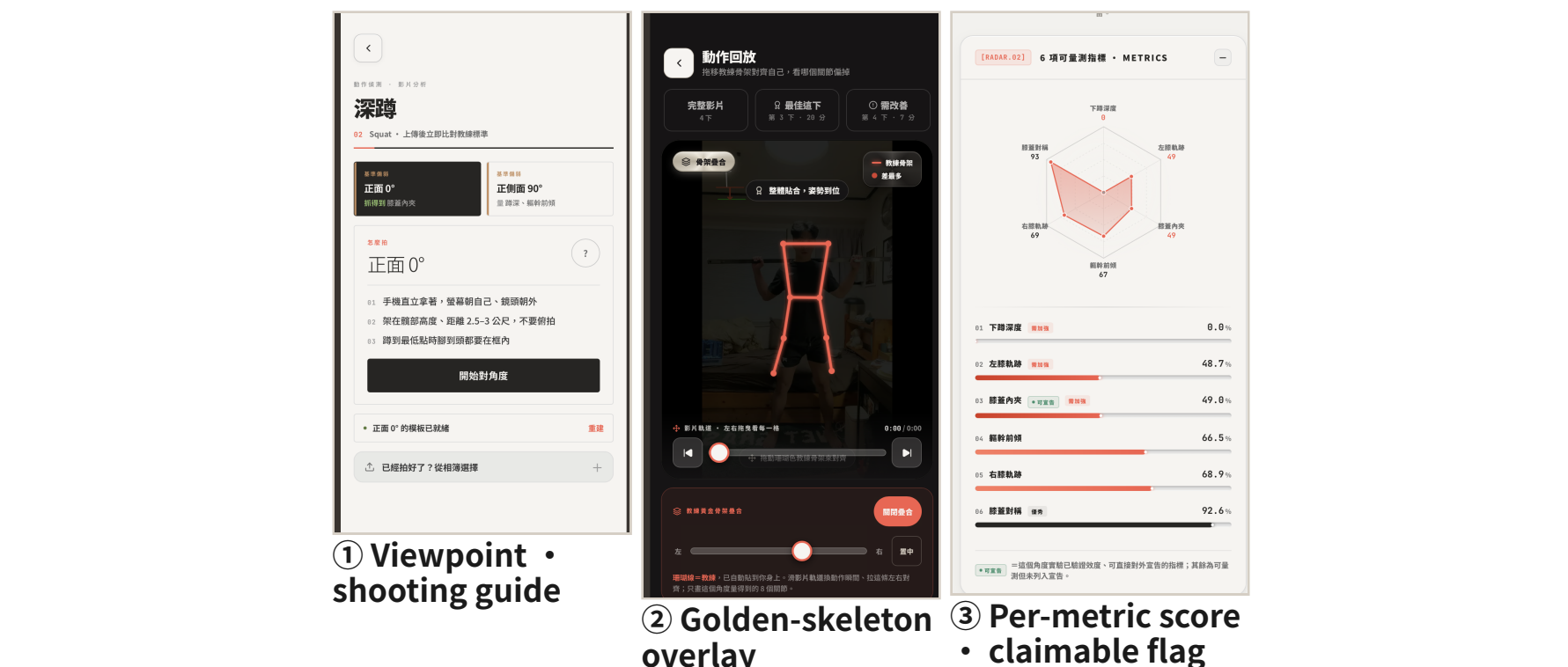


Fig. 9 Implemented as a working iOS prototype (device screenshots, Traditional-Chinese UI). The conclusions of the three-stage validation are surfaced directly in the interface: the system first declares what it can measure at that viewpoint, and only then gives a score.

Capability bounded by evidence

Wrongly declaring a fault "detectable" can create a **false sense of safety**, so the system adopts a **conservative capability-claim policy**: only viewpoints that passed empirical validation are listed as validated detection capability; everything else is labelled outside the validated scope.

